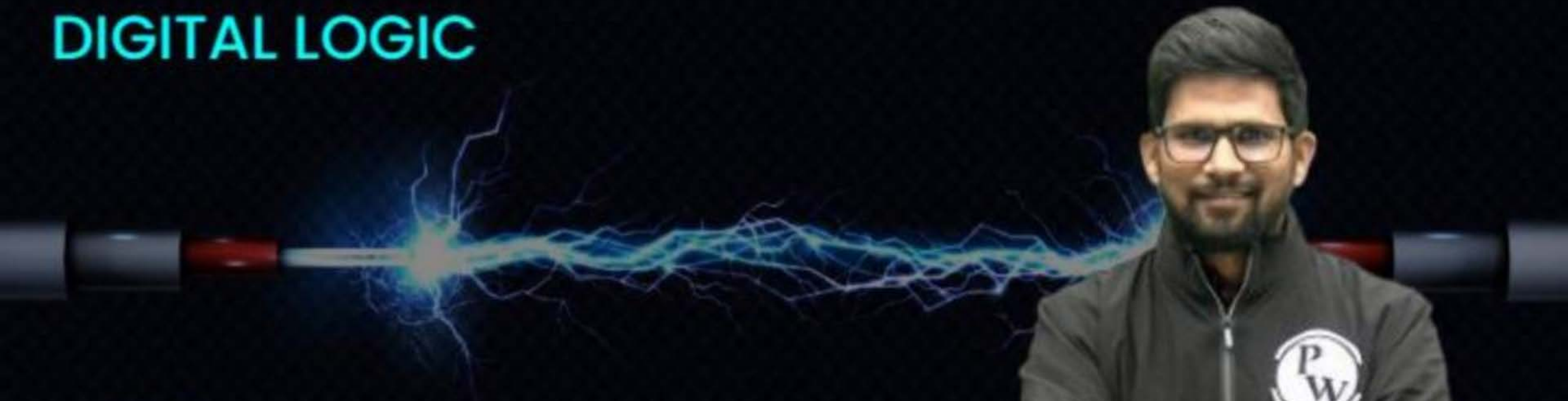
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 05

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture



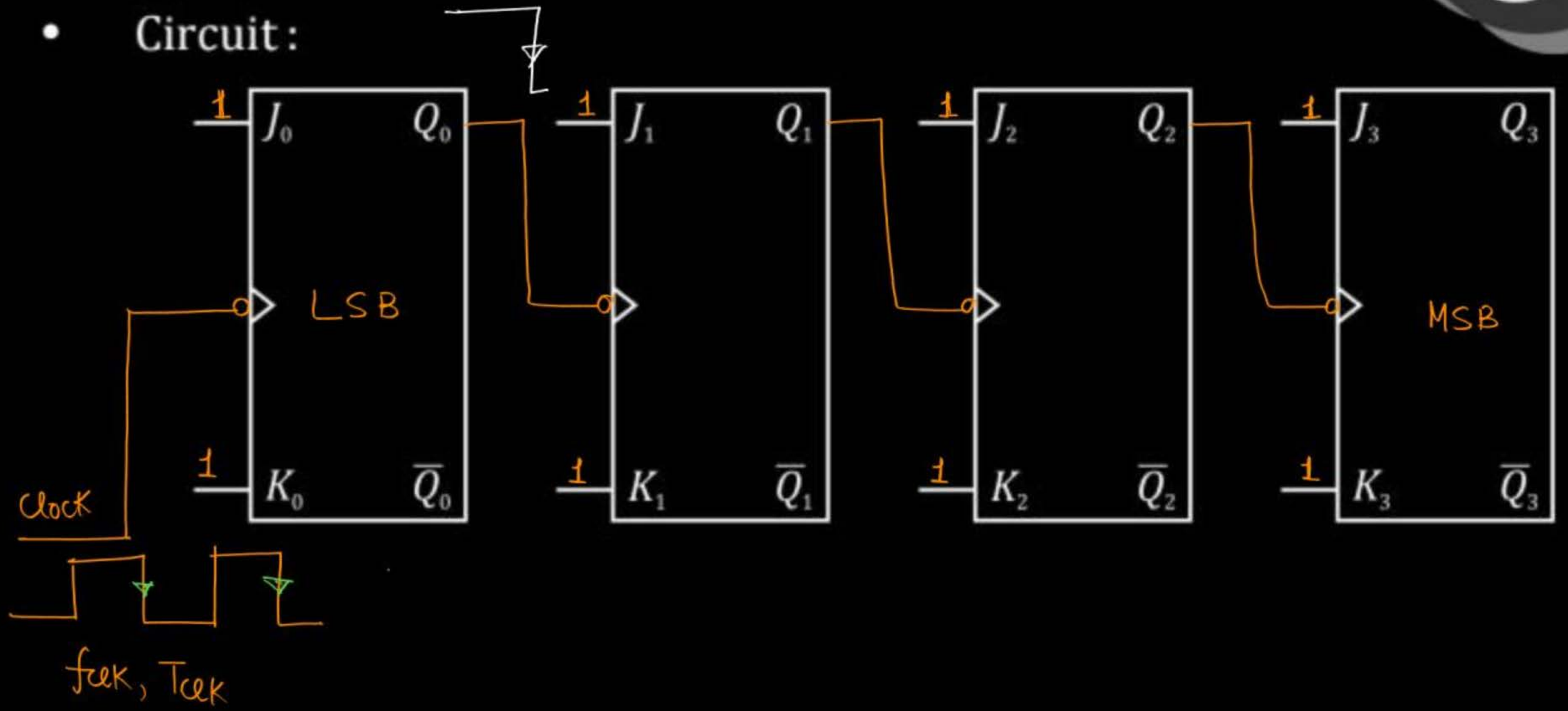
- Asyn. Counter
-
-



Asyn. Counter Cont.

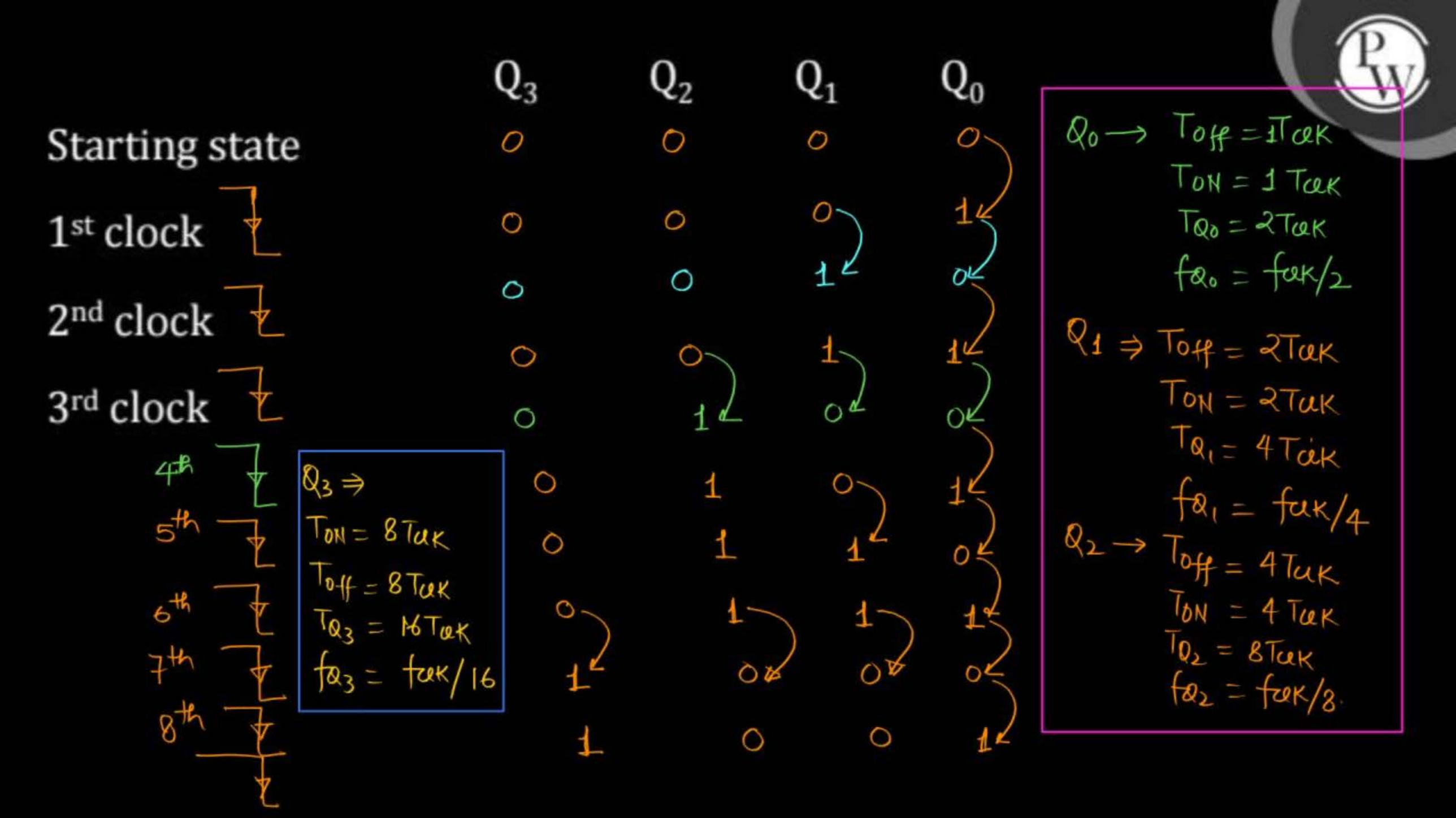
4-bit Asynchronous Counter

- Circuit :



- Working :

- Q_0 toggles in every clock.
- Q_1 toggles when Q_0 take transition from (1-0).
- Q_2 toggles when Q_1 take transition from (1-0)
- Q_3 toggles when Q_2 take transition from (1-0)


Q3
Q2
Q1
Q0
Starting state
1st clock
2nd clock
3rd clock
4th
5th
6th
7th
8th
Q3 ⇒
TON = 8 Tclk
Toff = 8 Tclk
TQ3 = 16 Tclk
fQ3 = fclk/16
Q0 → Toff = 1 Tclk
TON = 1 Tclk
TQ0 = 2 Tclk
fQ0 = fclk/2
Q1 ⇒ Toff = 2 Tclk
TON = 2 Tclk
TQ1 = 4 Tclk
fQ1 = fclk/4
Q2 → Toff = 4 Tclk
TON = 4 Tclk
TQ2 = 8 Tclk
fQ2 = fclk/8

10th 1 0 1 0

11th 1 0 1 1

12th 1 1 0 0

13th 1 1 0 1

14th 1 1 1 0

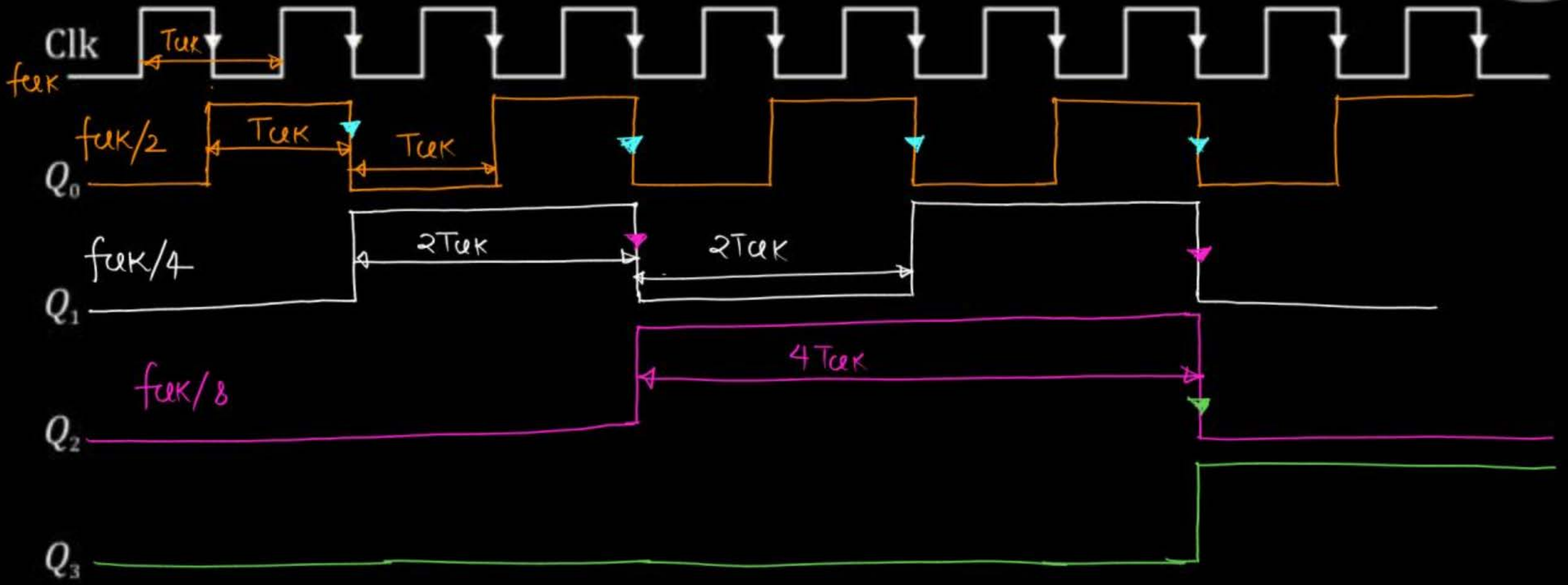
15th 1 1 1 1

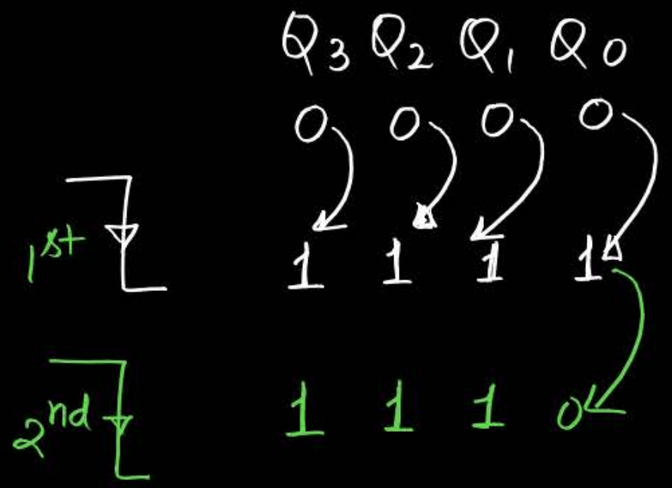


16th clock (0 0 0 0)

MOD No. = 16

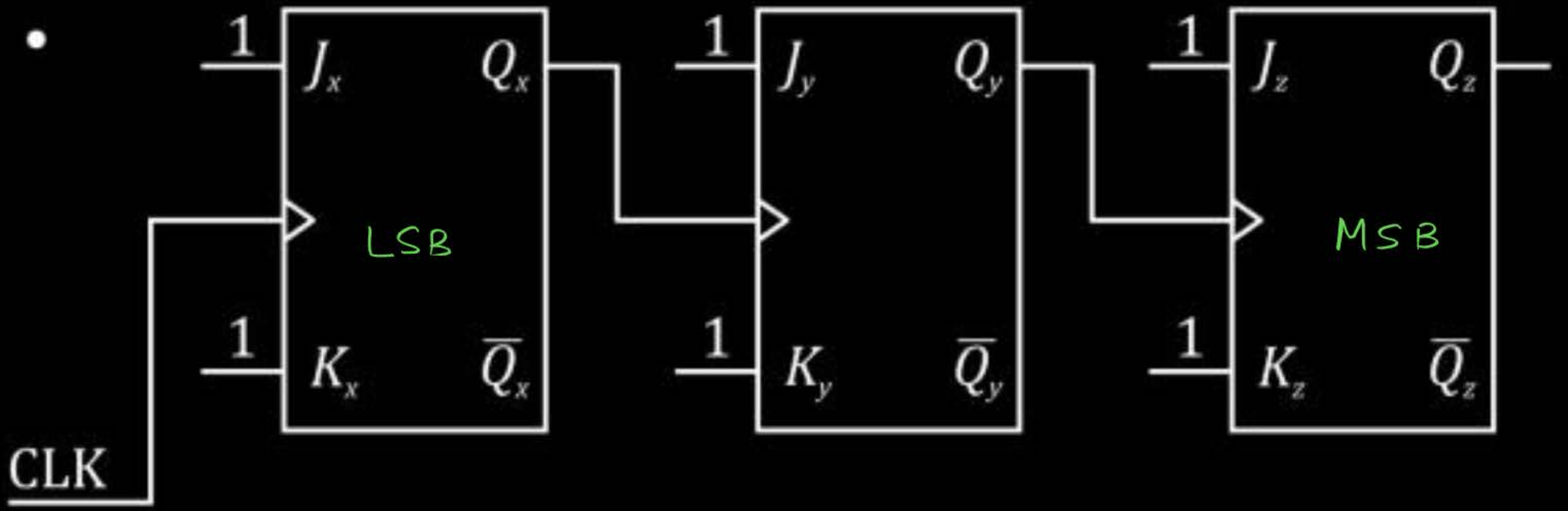
- Clock Diagram :





down Counter

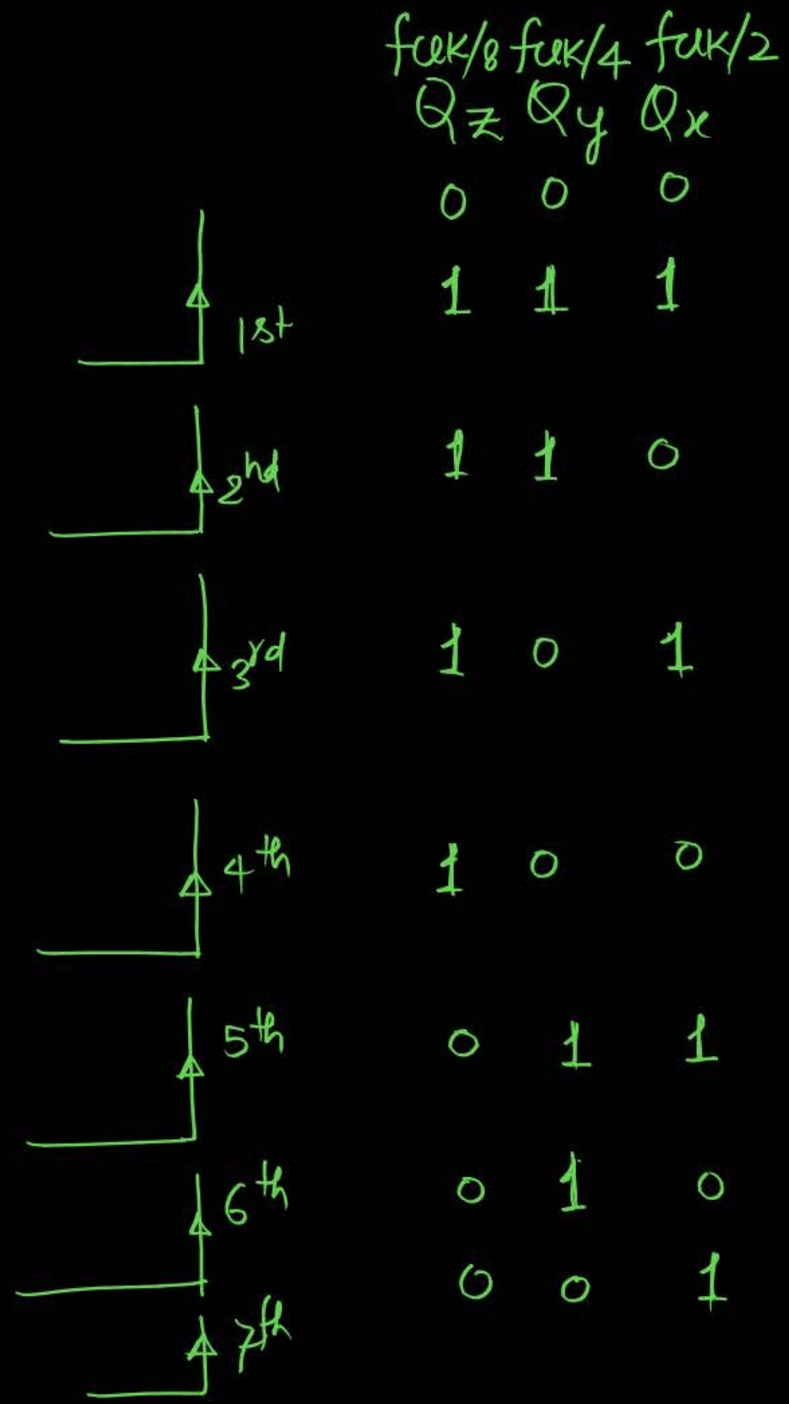
How to identify LSB bit and type (up or down) of Asynch. Counter



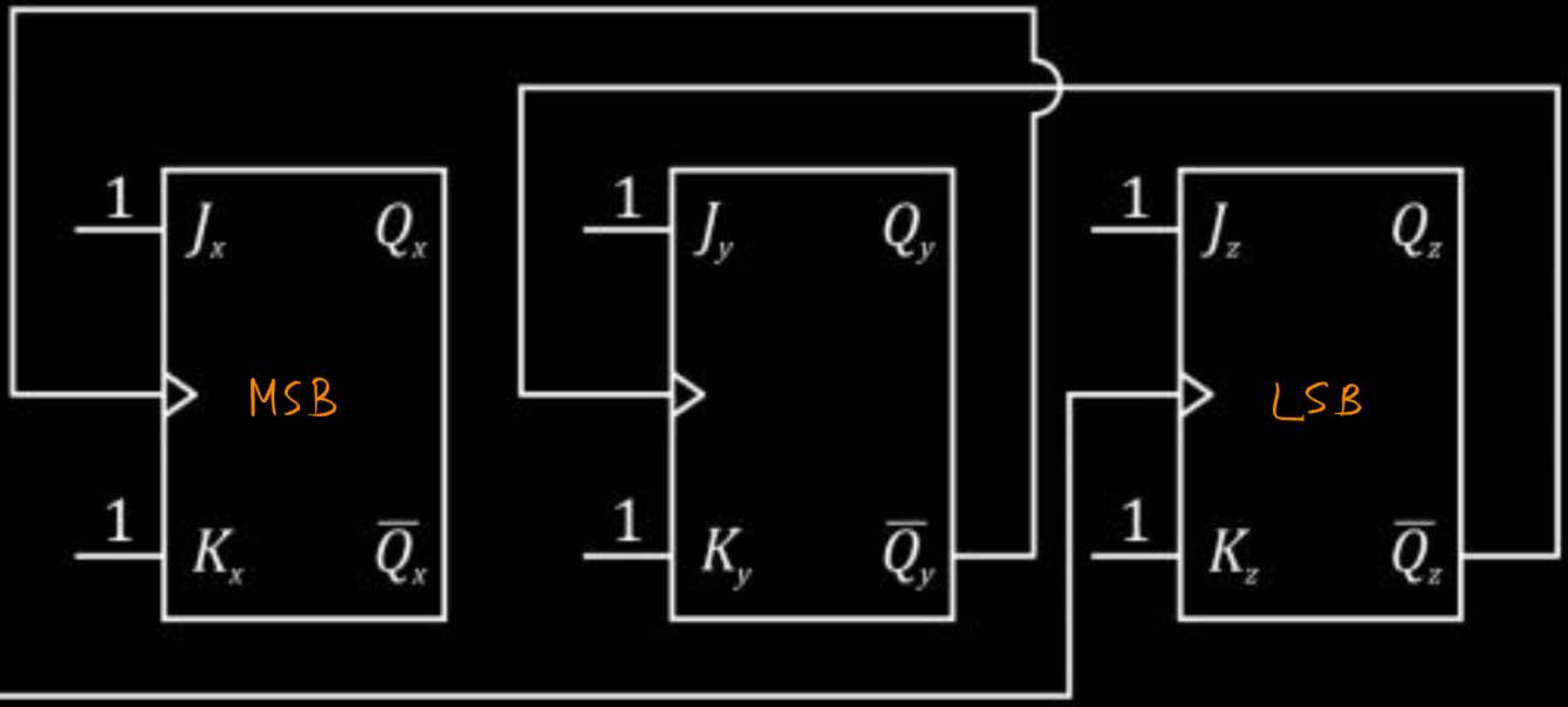
LSB Bit → Q_x

MSB Bit → Q_z

Up counter or down counter ? → down Counter.


fclk/8 fclk/4 fclk/2
Qz Qy Qx
0 0 0
1st
1 1 1
2nd
1 1 0
3rd
1 0 1
4th
1 0 0
5th
0 1 1
6th
0 1 0
7th
0 0 1

MOD - 8

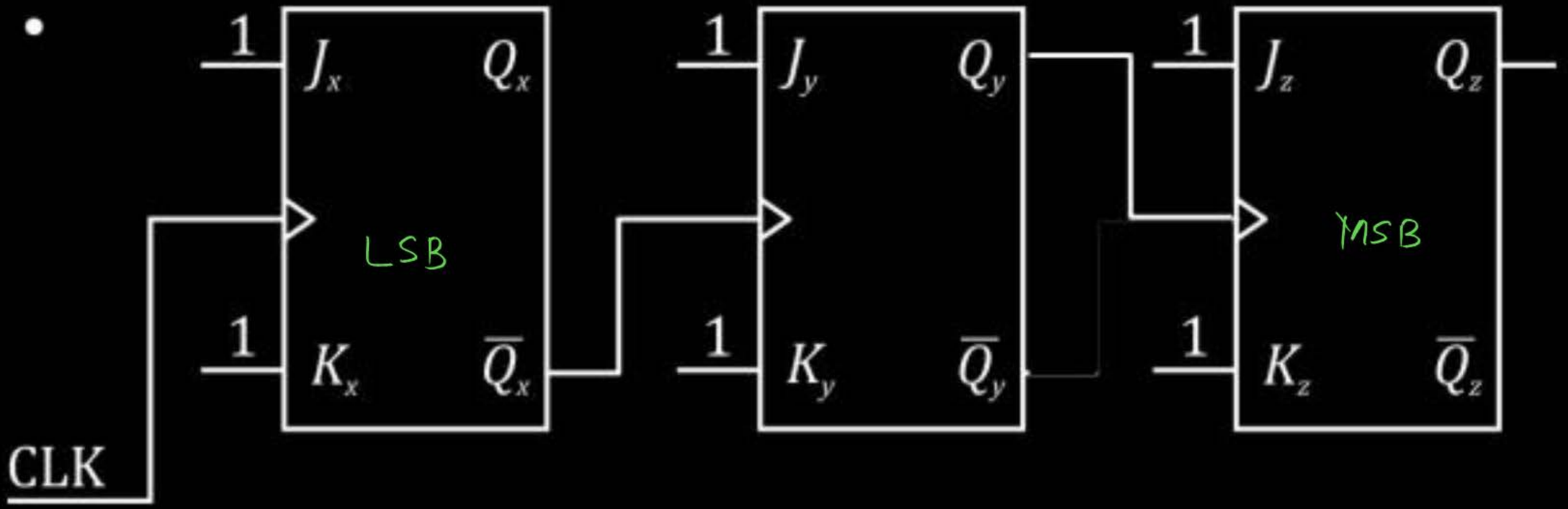


MOD – 8

LSB Bit → Q_z

MSB Bit → Q_x

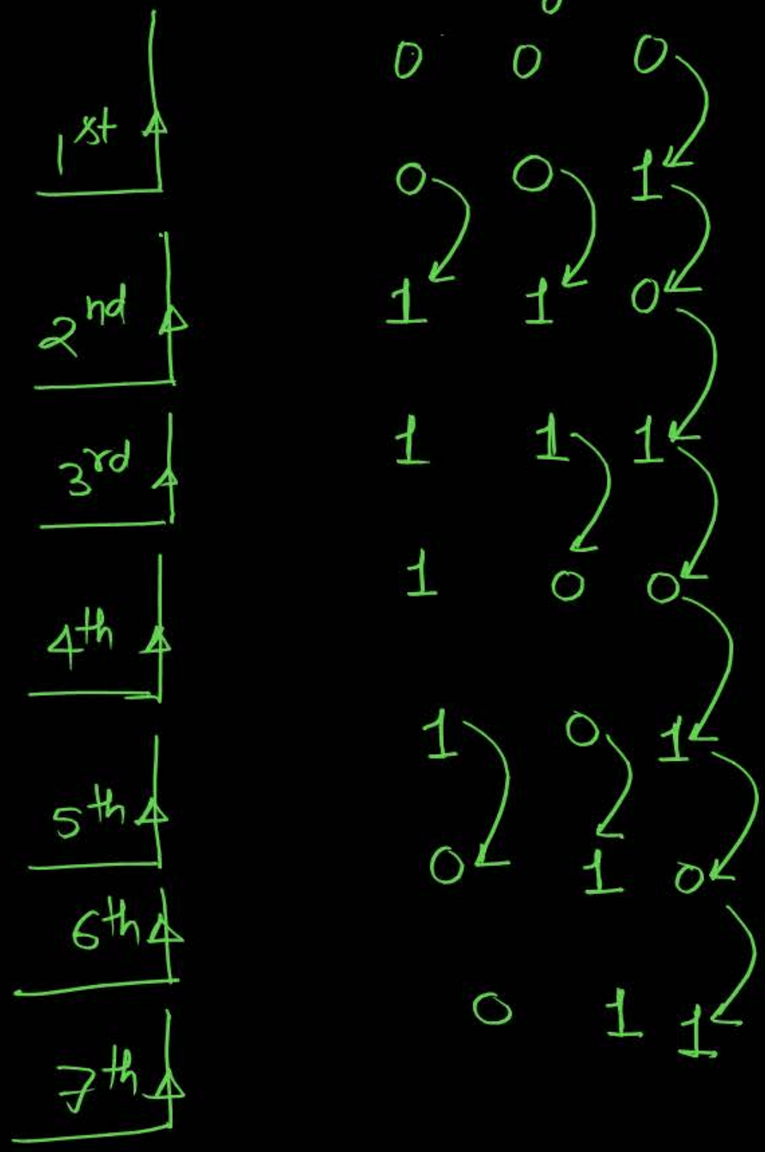
Up counter or down counter ? → Up counter



What is the MOD no. of above counter ?

Sequence generated is ?

$$\begin{pmatrix} 0-1-6-7-4 \\ -5-2-3 \end{pmatrix}$$





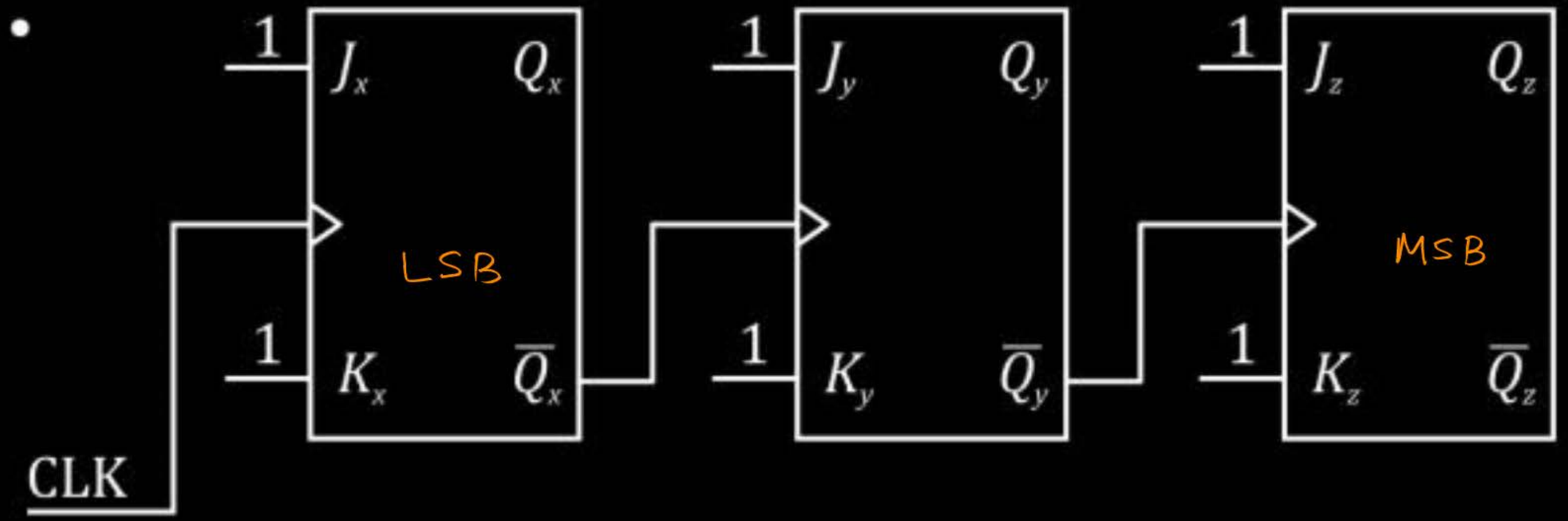
MOD – 8 Counter

$f_x = ?$ $f_{CLK}/2$ $f_y = ?$ $f_{CLK}/4$

$f_z = ?$ $f_{CLK}/8$

- If starting state is $Q_zQ_yQ_x = (100)_2$ then after 11 CLK pulses counter will be at state – $(3)_{10}$.

$(100)_2 \xrightarrow{8\,CLK} (100)_2 \xrightarrow{9^{th}} (5)_{10} \xrightarrow{10^{th}} (2)_{10} \xrightarrow{11^{th}} (3)_{10}$



LSB Bit → Q_x

MSB Bit → Q_z

up counter.

- Final Conclusion :

Trigger type	Clock i/p from previous o/p	Counter type
+ve edge trigger	Q	down counter
+ve " "	$\overline{Q}$	up counter
−ve " "	Q	up counter
−ve " "	$\overline{Q}$	down counter

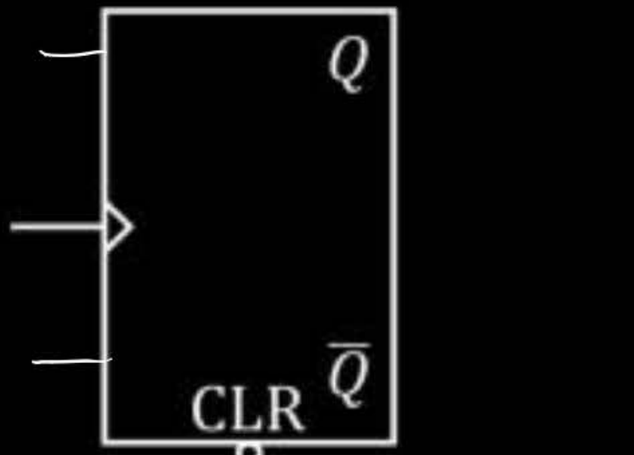
→ down Counter

Clear and Preset input

Clear input or CLR PIN input : This pin is asynchronous in nature and when activated, will reset (clear) the o/p of ff irrespective of the clock i/p or other i/ps.

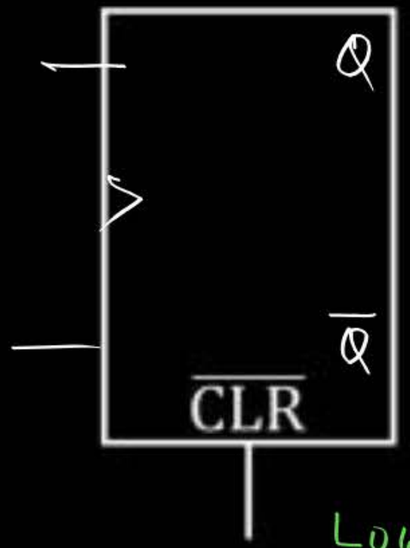
Preset input or PRE PIN input : This pin is asynchronous in nature and when activated, will set the o/p of ff irrespective of the clock i/p or other i/ps.

Low Enable and High Enable CLR and PRE input :

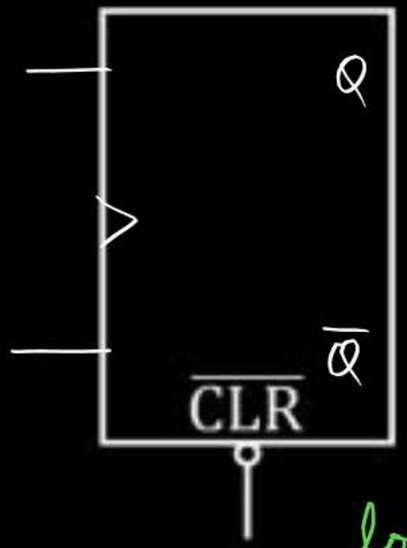


Low enable

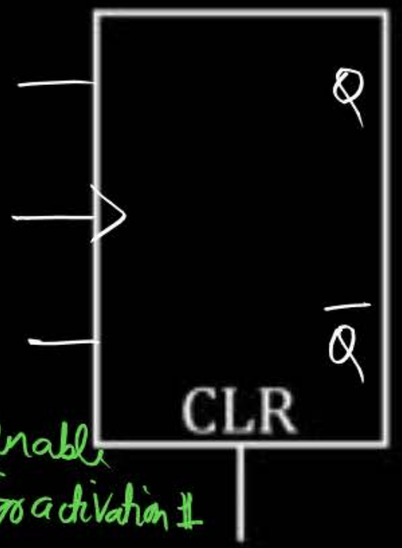
0 for activation



Low enable

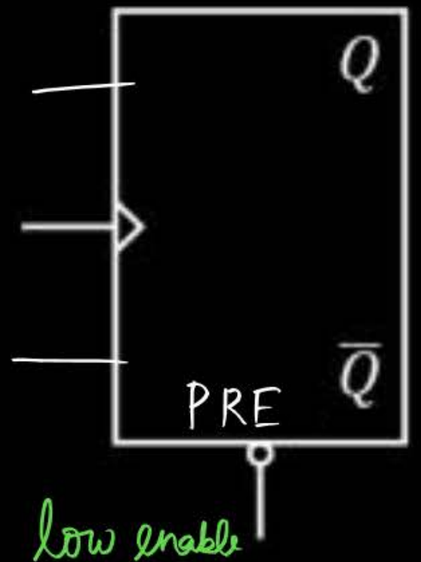


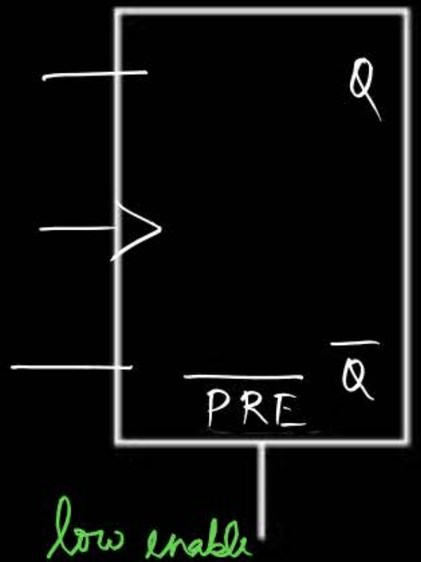
low enable

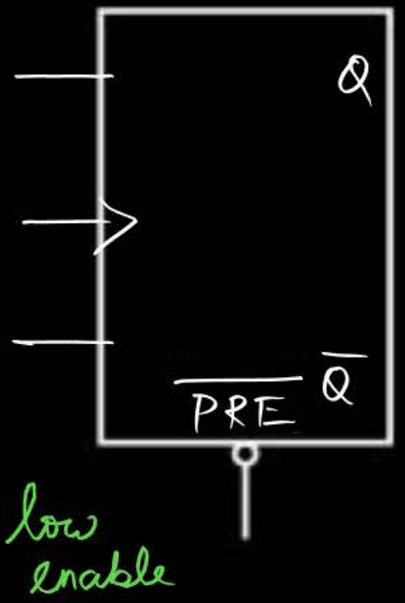


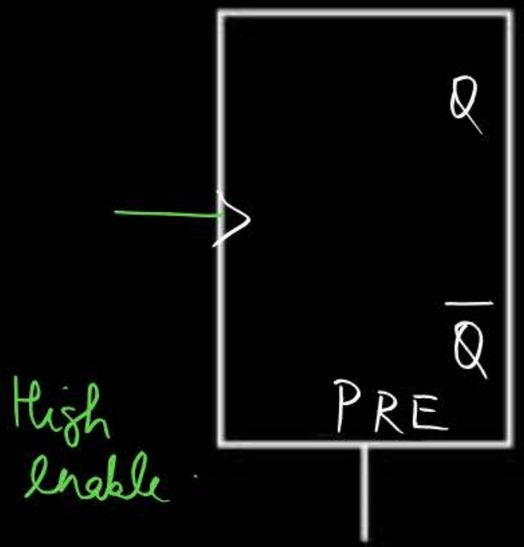
High enable for activation

Low Enable and High Enable CLR and PRE input :







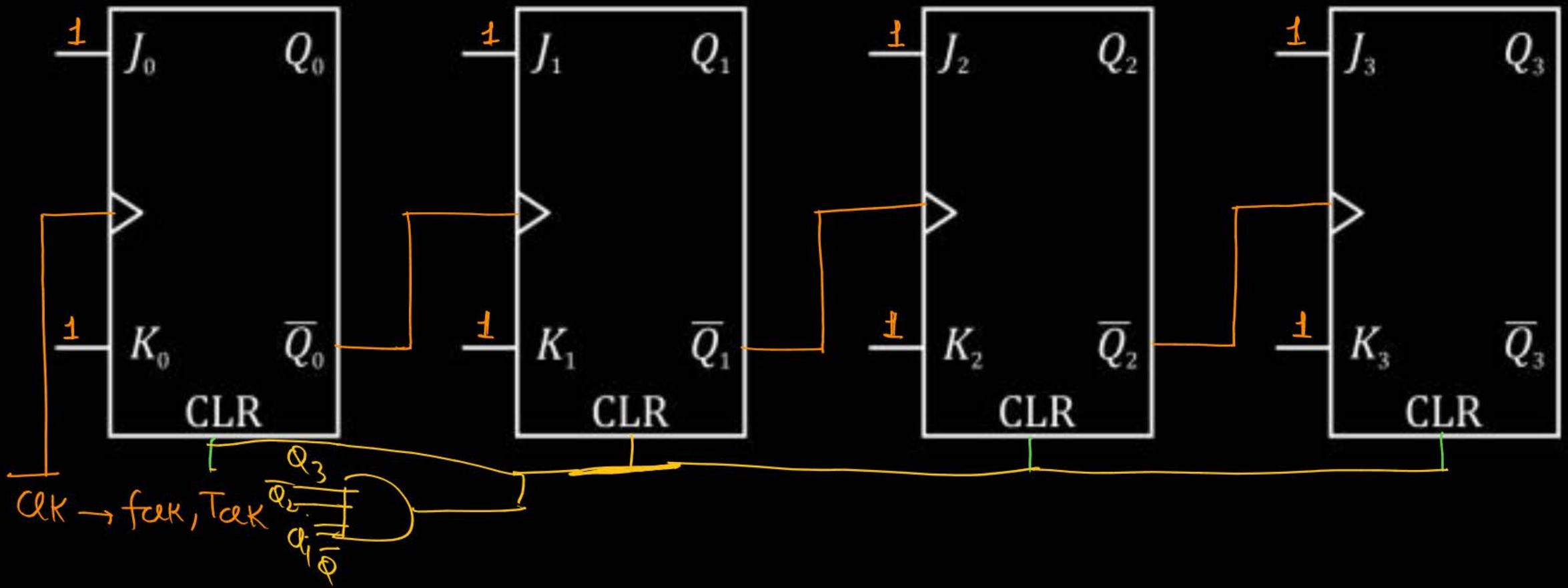


Designing of Asynchronous Counter with MOD no. other than 2^n

- BCD Counter Design :

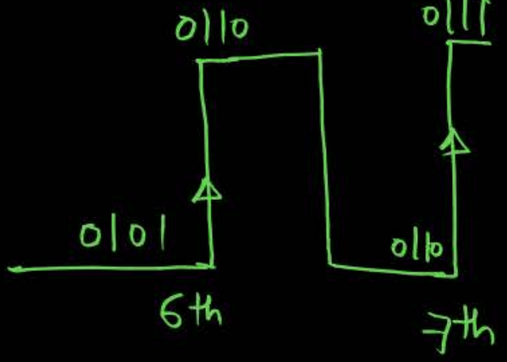
  BCD Counter Counts → (0 – 1 – 2 – 3 – 4 – 5 – 6 – 7 – 8 – 9) → MOD no. = 10

- Design :

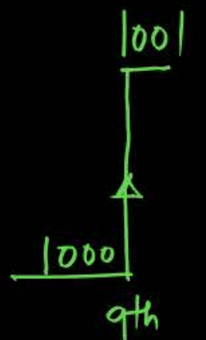


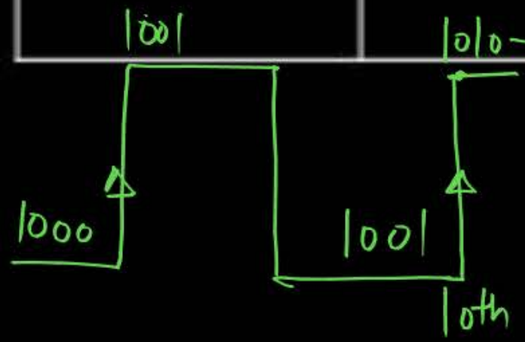
- Table :

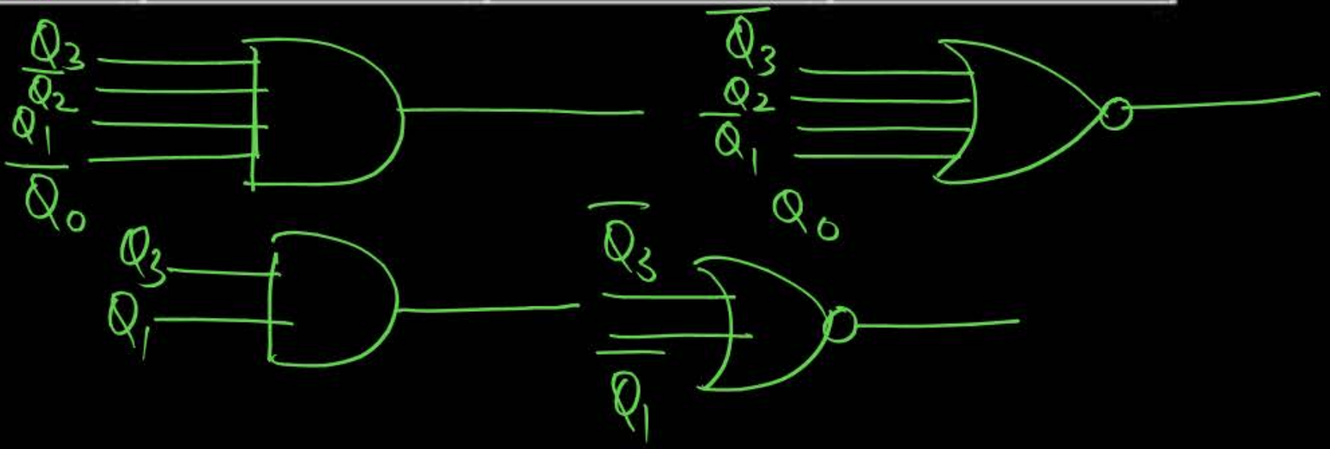
CLK	Q_3	Q_2	Q_1	Q_0
Start	0	0	0	0
1st ↑	0	0	0	1
2nd ↑	0	0	1	0
3rd ↑	0	0	1	1
4th ↑	0	1	0	0
5th ↑	0	1	0	1
6th ↑	0	1	1	0
7th ↑	0	1	1	1
8th ↑	1	0	0	0



CLK	Q_3	Q_2	Q_1	Q_0
9th ↑	1	0	0	1
10th ↑	1	0	1	0
		→ will exist momentarily and		
		will not be counted in MOD no.		







- **IMP Points :**

- BCD Counter is a MOD–10 Counter but not all MOD–10 counters are BCD Counter.

H.W: (0 – 1 – 2 ..... 11)

H.W: (7 – 6 – 5 – 4 – 3)

H.W: (0 – 7 – 6 – 5 – 4 – 3 – 2)

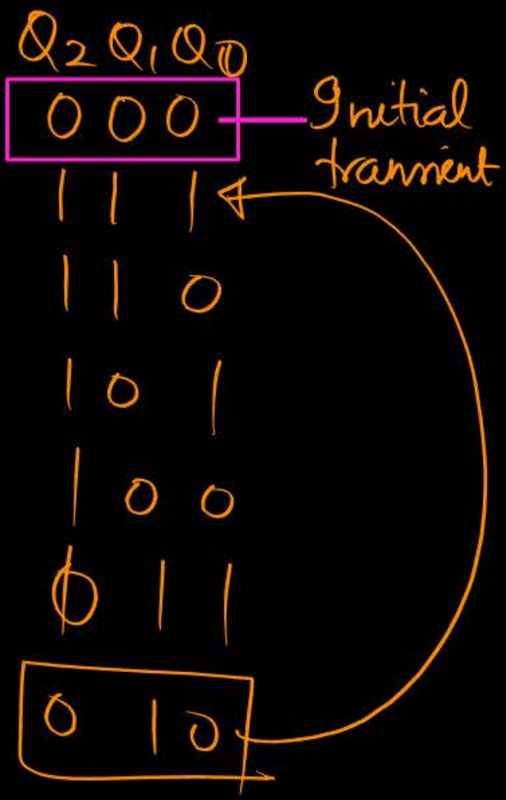
H.W: (7 – 0 – 1 – 2 – 3)

H.W: (1 – 2 – 3 – 4 – 5 – 6)

7 – 6 – 5 – 4 – 3 – 2

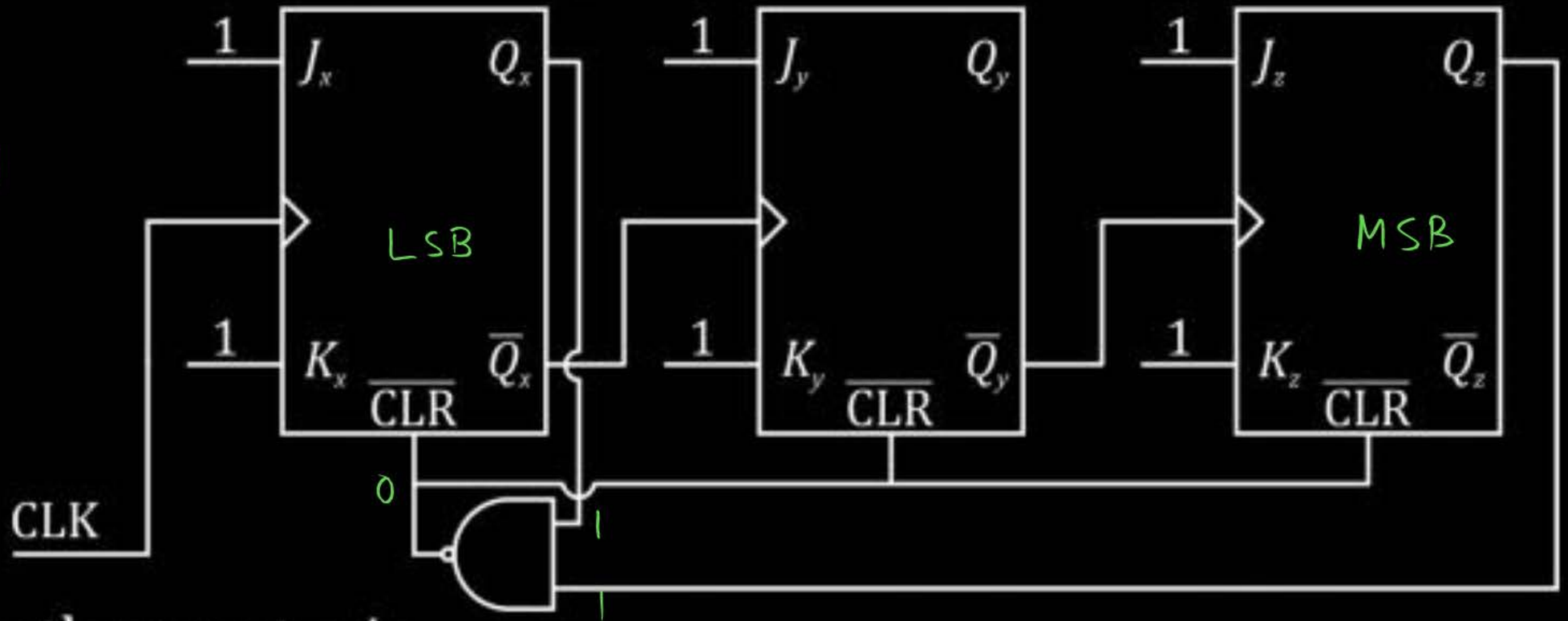
(111)

PRESET



Question

A Counter is designed as given below :



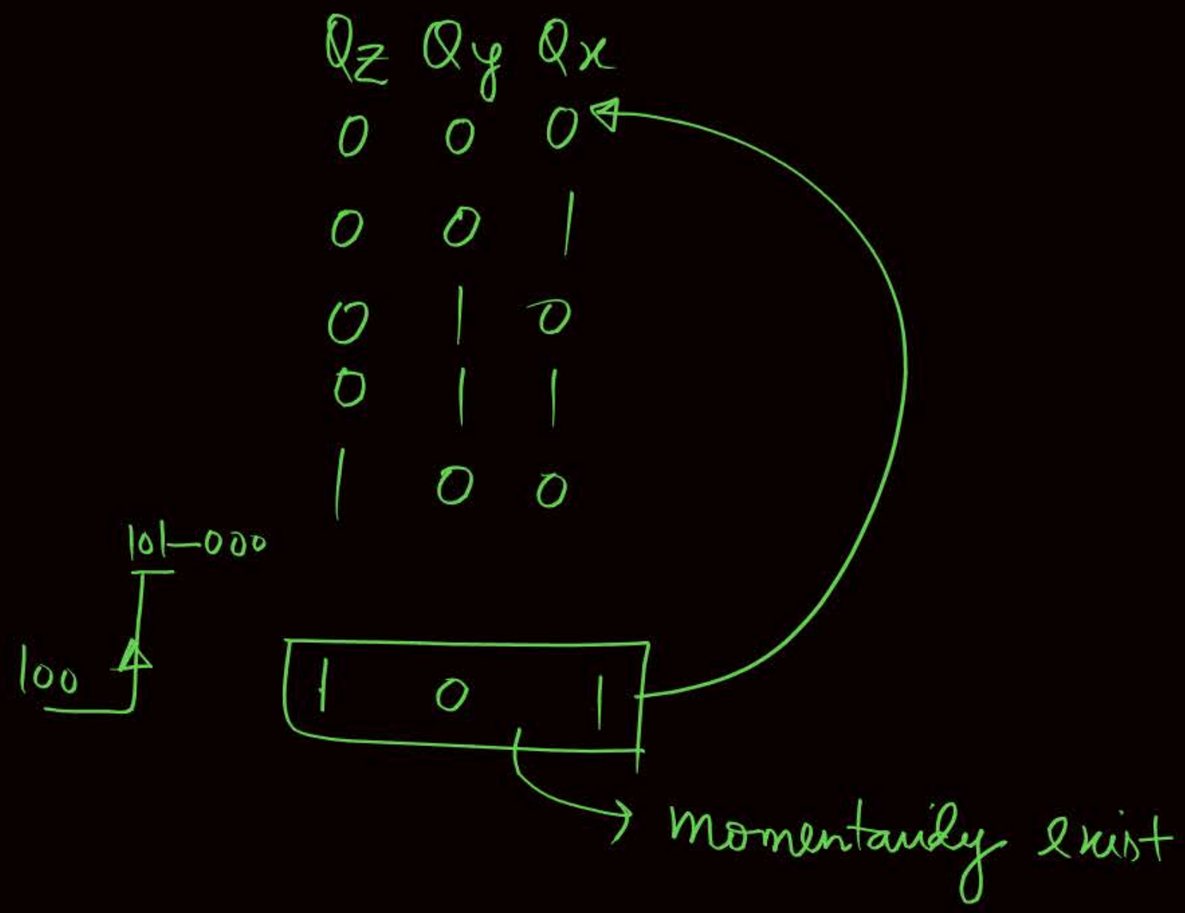
Then the counter is

(a) MOD-6 up counter

(b) MOD-5 down counter

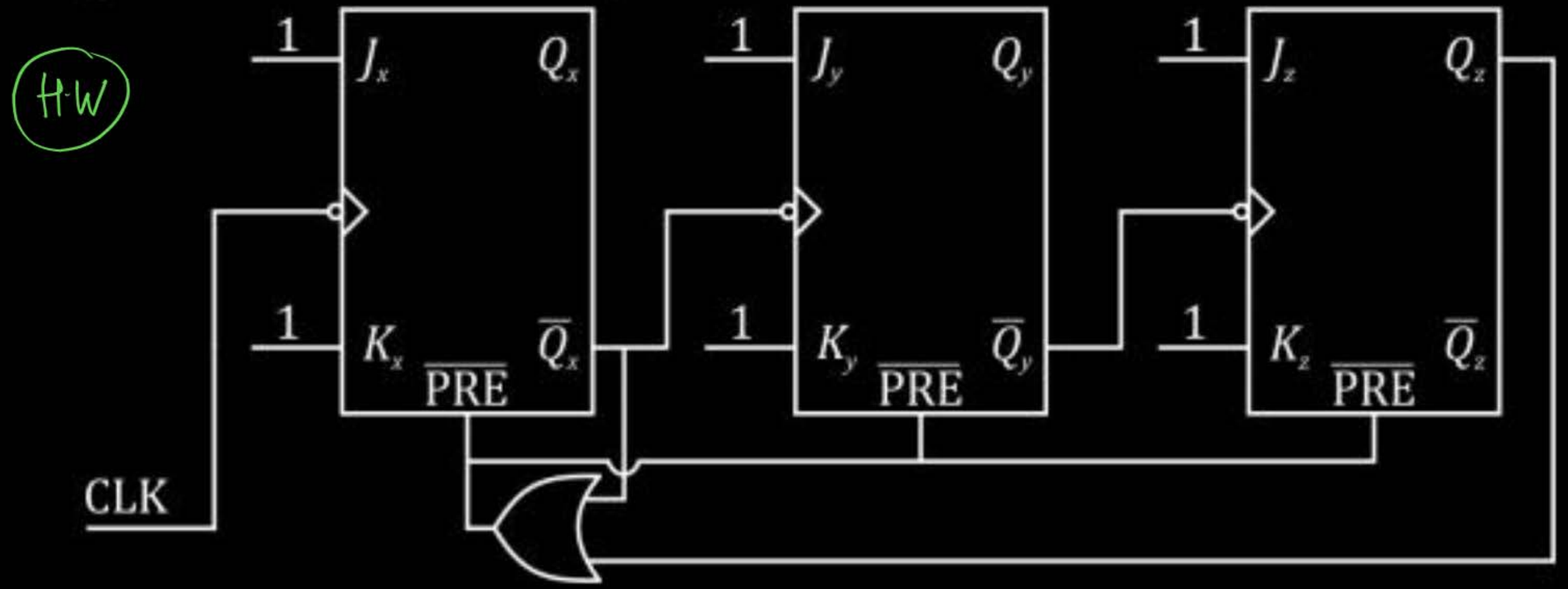
(c) MOD-5 up counter

(d) MOD-6 down counter


Qz Qy Qx
0 0 0
0 0 1
0 1 0
0 1 1
1 0 0
101—000
100
1 0 1
momentarily exist

Question

A sequential circuit is as given below :



The circuit works as :

(a) MOD-5 down counter

(b) MOD-4 down counter

(c) MOD-6 down counter

(d) MOD-5 up counter



Topic : 2 Min Summary

Asyn. Counter designing.

Thank you

GW

Soldiers !

